Fundamental of Mathematics

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Set Theory

Definition of Set:

By well defined we mean there is a rule by which one can decide whether an element belongs to the set or not.

Ex: **(i)** Natural numbers, prime numbers, real numbers less than 10, all are well defined objects.

(ii) But collection of good students in a class, collection of good batsmen are not sets because good student and good batsman are not well defined.

Definition

A set is a collection of well-defined objects which are distinct from each other. Sets are generally represented by capital letters A, B, C, ... etc and the elements of the set by a, b, c, ... etc.

Some Important Number Sets:

N = Set of natural numbers = {1, 2, 3, 4, ...}
W = Set of whole numbers = {0, 1, 2, 3, ...}
Z or **I** = Set of all integers = {0, ±1, ±2, ±3, ...}
$\mathbf{Z^+}$ = Set of positive integers = {1, 2, 3, ...} =N
$\mathbf{Z^-}$ = Set of negative integers = {−1, −2, −3, ...}
$\mathbf{Z_0}$ = Set of all non-zero integers = {±1, ±2, ±3, ...}
Q = Set of all rational numbers = {p/q; p, q ∈ I, q ≠ 0}
R = Set of all real numbers.
R−Q = Set of all irrational numbers Ex. $\{\sqrt{2}, \sqrt{3}, \pi, e, \log 2 .....\}$

Note: Symbol "∈" denotes "belongs to"

Methods of writing a set:

1. Roster Method:

In this method a set is described by listing elements, separated by commas and enclosing them by curly brackets. It is also called **Tabular Method**.

Ex: The set of vowels = {a, e, i, o, u}

2. Set builder Method:

In this case we write down property or rule "p" which gives us all the elements of the set.
A = {x: p(x)}

Ex: A = {x, x ∈ N and x = 2n for n ∈ N} = {2, 4, 6, ...}
B= {x^2, x ∈ Z} = {0, 1, 4, 9, ...}

Q.1 Write the following sets in set builder form
A = {1, 4, 9, 16, ...}
B = {2, 9, 28, 65, ...}
C = {13, 17, 19, 23, 29, 31}

Sol. A = {x : x = n^2, n ∈ N} or {x^2 ; x ∈ N}
B = {x : x = n^3 + 1, n ∈ N} or {x^3 + 1 ; x ∈ N}
C = {x : x is prime number, 13 ≤ x ≤ 31}

Q.2 Write the following sets into tabular form.
A = {x : x^2 − 4 = 0, x ∈ N}
B = {x : x = 2 y, 1 ≤ y ≤ 3, x ∈ N}

Sol. For A ⇒ x^2 − 4 = 0
⇒ (x + 2) (x − 2) = 0
x = 2, x = −2 (non-natural number)
∴ A = {2}
For B ⇒ ∵ x = 2y and 1 ≤ y ≤ 3
∴ 2 ≤ x ≤ 6 and x ∈ N
∴ B = {2, 3, 4, 5, 6}

Types of sets

1. Null set or Empty set:

A set having no element is called null Set or Empty set or void set. It is denoted by ϕ or { }.

Ex: A = {x ∈ N; 5 < x < 6} = ϕ
B = {x: x is a prime number, x < 2}
The sets which have atleast one element are called non-empty / non-void sets.

2. Singleton set:

A set consisting of a single element is called a singleton set.

Ex: A = {0}

3. Finite set:

A set which has finite number of elements is called finite set.

Ex: A = {a, b, c}, B = {1}, C = { }

Order of a finite set A is the number of elements in the set. It is denoted by **n(A)** or **O(A)** It is also called **cardinal number** of set A.

Ex: If A = {1, 2, 3}, then n(A) = 3

4. Infinite set:

A set which has infinite number of elements is called infinite set.

Ex: A = {1, 2, 3, 4, 5, ...}

5. Equal sets:

Two sets A & B are called equal if every element of set A is an element of B and every element of set B is an element of set A. If A and B are equal sets, then we write A = B and if they are unequal, then we write A ≠ B.

Ex:

A = {1, 2, 4} and B = {x: $x = 2^y$, y = 0, 1, 2} are equal sets.

6. Equivalent sets:

Two finite sets A and B are equivalent if they have same number of elements i.e. n(A) = n(B)

Ex:

A = {1, 2, 3] and B = {4, 5, 6} because n(A) = n(B) = 3

Point to Remember!!!

If two sets are equal, then they are equivalent, but converse is not always true.

SUBSETS & SUPERSETS :

If every element of set A is an element of set B, then A is subset of B and B is superset of set A, and we write $A \subseteq B$

Ex: If A = {5, 7} and B = {1, 5, 7, 9} then $A \subseteq B$
A is subset of B and B is superset of A

PROPER SUBSET :

If A is subset of set B such that $A \neq B$, then A is called proper subset of B and we write $A \subset B$

Ex: A = {1, 2}, B = {1, 2, 3}, C = {1, 2, 3}
A, B are subsets of C, A is proper subset of C but B is not proper subset of C (as B = C).

UNIVERSAL SET :

A set consisting of all possible elements which occur in the discussion is called a Universal set. It is denoted by "U".

Ex: If A = {1, 2}, B = {1, 3, 5}, C = {2, 5, 6} then U = {1, 2, 3, 4, 5, 6} can be taken as universal set.

POWER SET :

Set of all subsets of set A is called power set of A and it is denoted by P(A)

Ex: For A = {1, 2} power set P(A) = {ϕ, {1}, {2}, {1, 2}}

If n(A) = m, then n (P(A)) = 2^m

Point to Remember!!!

1. Every set is subset of itself. ($A \subseteq A$)
2. Null set (ϕ) is subset of every set ($\phi \subseteq A$)
3. $N \subset W \subset Z \subset Q \subset R \subset C$
4. If n(A) = m then total number of subsets of A = 2^m.
   Proper subsets of A = $2^m - 1$
5. $A \subseteq B$ and $B \subseteq A \Leftrightarrow A = B$

Q.1 Find the value of $n(P(P(P(\phi))))$

Sol. $\because n(\phi) = 0 \Rightarrow n\left(\underbrace{P(\phi)}_{B(let)}\right) = 2^0 = 1$

$\because n(B) = 1 \Rightarrow n\left(\underbrace{P(B)}_{C(let)}\right) = 2^1 = 2$

$\because n(C) = 2 \Rightarrow n(P(C)) = 2^2 = 4$

Q.2 If $n(A) = 4$, then find number of subsets of $P(A)$.

Sol. $n(A) = 4 \Rightarrow n(P(A)) = 2^4 = 16$

Hence number of subsets of $P(A) = \mathbf{2^{16}}$

Q.3 State TRUE or FALSE

(i) $\phi = \{0\}$

(ii) $\{a, b\} \subseteq \{a, b, c\}$

(iii) $\{a, b\} \in \{a, b, c\}$

(iv) $\{a\} \in \{\{a\}, b\}$

(v) $\phi \subset \{1, 2\}$

(vi) $\phi \in \{\phi, \{1\}\}$

(vii) $x \in A, A \in B \Rightarrow x \in B$

(viii) $x \in A, A \not\subset B \Rightarrow x \notin B$

(ix) $A \subseteq B, x \notin B \Rightarrow x \notin A$

(x) $A \subseteq B, B \in C \Rightarrow A \in C$

Sol.

(i) False: $\{0\}$ is a singleton set & ϕ is null set.

(ii) True: Clearly $[a, b]$ is a subset of $\{a, b, c\}$

(iii) False: $\{a, b\}$ is a subset, it is not an element for $\{a, b, c\}$ although a, b are elements but $\{a, b\}$ cannot be treated as a, b

(iv) True: Clearly $\{a\}$ is an element of $\{\{a\}, b\}$

(v) True: ϕ is proper subset of $\{1, 2\}$

(vi) True: ϕ is clearly one of the element of $\{\phi, \{1\}\}$

(vii) False: Let $x = 1, A = \{1\}, B = \{\{1\}, 2\}$
here $x \in A, A \in B$ but $x \notin B$

(viii) False: Let $x = 1, A = \{1, 2\}, B = \{1\}$
here $x \in A, A \not\subset B$ but $x \in B$

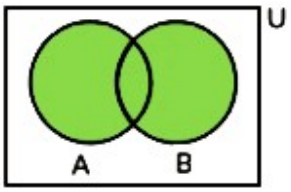
(ix) True: If $A \subseteq B$ then all elements of A lie in B and also, we know $x \notin B$ then it cannot be an element of A for sure.

(x) False: Let $A = \{1\}, B = \{1, 2\}, C = \{\{1, 2\}, 3\}$
Clearly $A \subseteq B$ and $B \in C$ but $A \notin C$

OPERATIONS ON SETS

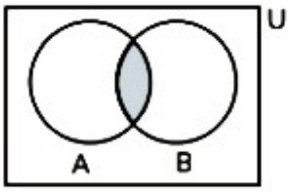
1. Union of two sets:

$A \cup B = \{x : x \in A \text{ or } x \in B\}$



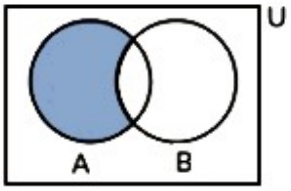
2. Intersection of two sets:

$A \cap B = \{x : x \in A \text{ and } x \in B\}$



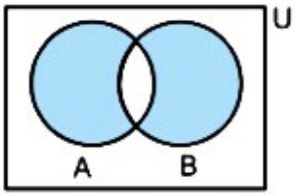
3. Difference of two sets:

$A - B = \{x : x \in A \text{ and } x \notin B\}$



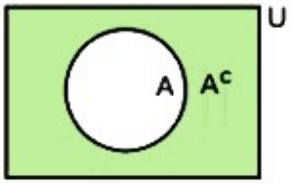
4. Symmetric difference of two sets:

$A \Delta B = (A-B) \cup (B-A)$



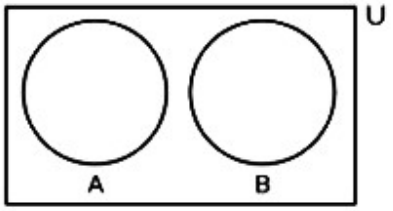
5. Complement of a set:

$\bar{A}$ or A' or $A^c = \{x : x \notin A \text{ and } x \in U\} = U - A$



6. Disjoint (Incompatible) sets:

Two sets A and B are called disjoint sets if they do not have any common element i.e $A \cap B = \phi$



Ex: If $U = \{1, 2, 3, 4, 5, 6, 7\}$ and $A = \{1, 2\}$, $B = \{2, 4, 5\}$
then $A \cup B = \{1, 2, 4, 5\}$
$A \cap B = \{2\}$
$A - B = \{1\}$, $B - A = \{4, 5\}$
$A^c = \{3, 4, 5, 6, 7\}$, $B^c = \{1, 3, 6, 7\}$
$A \Delta B = \{1, 4, 5\}$

Laws of Algebra of sets :

1. **Commutative Law :** $A \cup B = B \cup A$
   $A \cap B = B \cap A$
2. **Distributive Law :** $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
   $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
3. **Associative Law :** $(A \cup B) \cup C = A \cup (B \cup C)$
   $(A \cap B) \cap C = A \cap (B \cap C)$
4. **Idempotent Law :** $A \cup A = A$
   $A \cap A = A$
5. **Identity Law :** $A \cup \phi = A$ $\quad A \cup U = U$
   $A \cap \phi = \phi$ $\quad A \cap U = A$
6. **Demorgan's Law :** $(A \cup B)^c = A^c \cap B^c$
   $(A \cap B)^c = A^c \cup B^c$

Some Important Results :

1. $A - B = A \cap B^c$

2. $B - A = B \cap A^c$

3. $(A^c)^c = A$

4. $A \cap A^c = \phi$, $A \cup A^c = U$

5. $A-(B \cup C)=(A- B) \cap (A-C)$
$A-(B \cap C)=(A-B) \cup (A-C)$

6. $A \cap B \subseteq A$, $A \cap B \subseteq B$
$A \subseteq A \cup B$, $B \subseteq A \cup B$

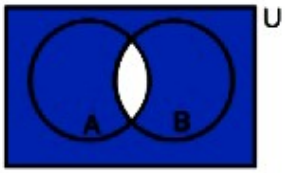
7. $A \subseteq B \Rightarrow A \cap B = A$
$A \subseteq B \Rightarrow A \cup B = B$

8. $A \subseteq B \Rightarrow B^c \subseteq A^c$

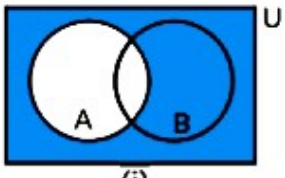
Q.1 Prove $(A \cap B)^c = A^c \cup B^c$ with and without Venn diagram.

Sol. **(i)** With Venn diagram.

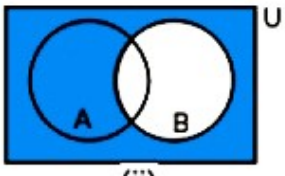
LHS



RHS

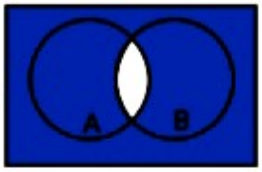


(i)



(ii)

(i) $\cup$ (ii) $\Rightarrow$



= LHS

(ii) Without using Venn diagram:

let $x \in (A \cap B)^c \Rightarrow x \notin (A \cap B)$

$\Rightarrow x \notin A$ or $x \notin B$

$\Rightarrow x \in A^c$ or $x \in B^c$

$\Rightarrow x \in A^c \cup B^c$

hence $(A \cap B)^c \subseteq A^c \cup B^c$ ..... (i)

Now let $y \in A^c \cup B^c$

$\Rightarrow y \in A^c$ or $y \in B^c$

$\Rightarrow y \notin A$ or $y \notin B$

$\Rightarrow y \notin A \cap B$

$\Rightarrow y \in (A \cap B)^c$

hence $A^c \cup B^c \subseteq (A \cap B)^c$ .....(ii)

By (i) & (ii)

$(A \cap B)^c = A^c \cup B^c$

Q.2 Prove the following using laws of algebra:

(i) $(A \cup B \cup C) \cap (A \cap B' \cap C')' \cap C' = B \cap C'$

(ii) $A' \cup \{(A \cup B) \cap B'\} = (A \cap B)'$

Sol. **(i) LHS** $= (A \cup B \cup C) \cap (A' \cup B \cup C) \cap C'$

$= \{A \cup (B \cup C) \cap \{A' \cup (B \cup C)\} \cap C'$

$= \{(A \cap A') \cup (B \cup C)\} \cap C'$

$= \{\phi \cup (B \cup C)\} \cap C'$

$= (B \cup C) \cap C'$

$= (B \cap C') \cup (C \cap C')$

$= (B \cap C') \cup (\phi) =$ **$B \cap C'$ = RHS**

(ii) LHS $= A' \cup \{(A \cap B') \cup (B \cap B')$

$= A' \cup \{(A \cap B'\} \cup \phi$

$= A' \cup (A \cap B')$

$= (A' \cup A) \cap (A' \cup B')$

$= U \cap (A' \cup B')$

$= A' \cup B' =$ **$(A \cap B)'$ = RHS**

THEOREMS ON CARDINAL NUMBERS

1. For two sets A & B

→ n (atleast one in A or B)

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

→ n(exactly one in A or B) $= n(A \cup B) - n(A \cap B)$

$= n(A) + n(B) - 2n(A \cap B)$

→ n (only A) = n(A − B)
= n(A) − n(A ∩ B)

→ n (neither A nor B) = n (A' ∩ B') = n(U) − n(A ∪ B)

2. For three sets A, B & C

→ n (atleast one in A, B or C)

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$$

$$n \text{ (atleast two in A, B or C)} = n(A \cap B) + n(B \cap C) + n(C \cap A) - 2\, n(A \cap B \cap C)$$

Q.1 If n(A) = 7, n(B) = 10, then find minimum and maximum value of n(A ∪ B)

Sol. ∵ $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$n(A \cup B)_{max} = n(A) + n(B) - n(A \cap B)_{min}$
$= 7 + 10 - 0$
$= 17$

$n(A \cup B)_{min} = n(A) + n(B) - n(A \cap B)_{max}$
$= 7 + 10 - 7$
$= \mathbf{10}$

Q.2 If n(A ∪ B) = 90, n(A − B) = 15, n(A ∩ B) = 30, then find n(B).

Sol. $n(A - B) = n(A) - n(A \cap B)$ ....(i)
$n(A \cup B) = n(A) + n(B) - n(A \cap B)$ ....(ii)
(ii) − (i)
$n(A \cup B) - n(A - B) = n(B)$
$90 - 15 = n(B) \Rightarrow n(B) = \mathbf{75}$

Q.3 In a group of 40 students 25 take tea, 17 take coffee and 8 take neither of the two. How many take both tea and coffee.

Sol. Let tea → T, coffee → C
given n(U) = 40, n(T) = 25, n(C) = 17, $n(T' \cap C') = 8$
$n(T' \cap C') = n\{(T \cup C)'\} = n(U) - n(T \cup C)$
$8 = n(U) - \{n(T) + n(C) - n(T \cap C)\}$
$8 = 40 - \{25 + (17) - n(T \cap C)\}$
$\Rightarrow n(T \cap C) = \mathbf{10}$

Q.4 A college awarded 38 medals in Football, 15 in basketball and 20 in cricket. If these medals went to a total of 58 players and only three players got medals in all the three sports, then how many players received medals in exactly two of the three sports.

Sol. Let Football → F, Basketball → B, Cricket → C

Given $n(F) = 38$, $n(B) = 15$, $n(C) = 20$

$n(F \cup B \cup C) = 58$, $n(F \cap B \cap C) = 3$

$n(F \cup B \cup C) = n(F) + n(B) + n(C) - \{n(F \cap B) + n(B \cap C) + n(C \cap F)\} + n(F \cap B \cap C)$

$58 = 38 + 15 + 20 - \{k\} + 3$

$\Rightarrow k = 18$

n (exactly two in F, B or C)

$= \{n(F \cap B) + n(B \cap C) + n(C \cap F)\} - 3\, n(F \cap B \cap C)$

$= k - 3(3) =$ **9**